The Standard Model on the world brane of 7+1 spacetime: symbols and their definitions

A dictionary of every symbol used for the Standard Model on the brane, electroweak symmetry breaking, quark and lepton mixing, dirac16complex at the brane, the portals and neutrino flavour conversion — sections 17–19 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation and its meaning, grouped by topic. Longer defining equations follow each table as numbered displayed formulas. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$, except where a unit is stated (GeV, eV, MeV, km, m, g/cm^3, K, Gyr).
- Coordinates $X = (x_0, \ldots, x_7)$: x_0 time, x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space; on the brane $x = (x_0, x_1, x_2, x_3)$ and $y = (x_4, \ldots, x_7)$.
- Indices: $\mu, \nu \in \{0, 1, 2, 3\}$ on the brane; $A, B \in \{0, \ldots, 7\}$ frame indices of 7+1 spacetime; $k, l \in \{4, \ldots, 7\}$ transverse; $i, j \in \{1, 2, 3\}$ generations or mass eigenstates; $\alpha, \beta \in \{e, \mu, \tau\}$ flavours; $a \in \{1, \ldots, 4\}$ transverse spinor index (sections 5–6) or colour index $a = 1, \ldots, 8$ of the gluons (section 2).
- Metrics: $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ in 7+1 dimensions (`η17`), simply the squares of the octonion basis ($\eta_{AB} = \mathrm{Re}(e_A e_B)$); $\eta = \mathrm{diag}(1, -1, -1, -1)$ on the brane, its restriction to $x_0, \ldots, x_3$: the squares of the basis $e_0 = 1, e_1, e_2, e_3$ of the quaternion subalgebra $\mathbb{H}$ of the octonions.
- Signature: the bulk has one time-like and seven space-like dimensions (in xAct the list $\{7, 1, 0\}$), the brane one time-like and three space-like dimensions (in the same counting $\{3, 1, 0\}$).
- Matrices: 1_n identity, † conjugate transpose, * complex conjugate, $\otimes$ Kronecker product (`KroneckerProduct`).

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Spacetime, brane and gauge fields

symbol	Mathematica	definition	meaning
W	—	$\{x_4 = \cdots = x_7 = 0\}$	world brane (3+1 dimensions)
T^4, L_c	—	$x_k \simeq x_k + L_c$	auxiliary torus and its coordinate period
$h_{\mu\nu}$	—	$g_{\mu\nu}(x, y = 0)$	induced metric of the brane
b_0	`b0K`	$b = b_0$ (modern epoch)	static auxiliary scale factor
$V_\perp$	`vPerp` (number), `vP` (symbol)	$b_0^4 L_c^4$	proper volume of the auxiliary torus
G_8, G_4	—	$G_4 = G_8/V_\perp$	Newton constants in 8 and 4 dimensions
κ, κ_8	`k8`	$8\pi G_8$	gravitational coupling
M_8, $M_{\rm Pl}$	—	$8\pi G_8 = M_8^{-6}$, $M_{\rm Pl}^2 = M_8^6 V_\perp$	reduced Planck masses ($M_{\rm Pl} = 2.435 \times 10^{18}$ GeV)
G_μ^a, W_μ^I, B_μ	`wW1`, `wW2`, `wW3`, `bB` (for W^I, B)	—	gauge fields of $SU(3)$, $SU(2)$, $U(1)_Y$
g_s, g, g'	`gW`, `gY` (symbolic); `gEW`, `gpEW`	(2.2)	gauge couplings
T^a, f^{abc}	—	$[T^a, T^b] = i f^{abc} T^c$	$SU(3)$ generators and structure constants
τ^I, σ_i	`tauP`, `PauliMatrix[i]`	Pauli matrices	$SU(2)$ generators $\tau^I/2$
D_μ	—	(2.1)	gauge- and gravity-covariant derivative on the brane
$\gamma^\mu(x)$	—	$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2h^{\mu\nu}$	Dirac matrices of the curved brane

$$D_\mu = \nabla_\mu - i g_s\, G_\mu^a T^a - i g\, W_\mu^I \tfrac{\tau^I}{2} - i g'\, Y\, B_\mu, \tag{2.1}$$

$$g = \frac{2m_W}{v}, \qquad \cos\theta_W = \frac{m_W}{m_Z}, \qquad g' = g\tan\theta_W, \qquad e = g\sin\theta_W. \tag{2.2}$$

3. Matter, Higgs and electroweak quantities

symbol	Mathematica	definition	meaning
Q_L, u_R, d_R, L_L, e_R, ν_R	`smF`	quantum numbers (N_c, d, Y, χ)	quark and lepton fields of one generation
χ_f	`smF[[i,5]]`	$+1$ left-handed, -1 right-handed	chirality sign in the anomaly sums
Y, T_3, Q	—	$Q = T_3 + Y$	hypercharge, weak isospin, electric charge
$A_{\dots}$	`anom`	(3.1)	anomaly coefficients
g_*	`gStar`	$28 + \frac{7}{8}\cdot 90$	relativistic degrees of freedom (106.75)
Φ, $\tilde\Phi$	—	$\tilde\Phi = i\sigma_2\Phi^*$	Higgs doublet and its conjugate
μ^2, λ	`lamH`	$\lambda = m_H^2/(2v^2)$	Higgs potential parameters
v	`vEW`, `vH`	$(\sqrt{2}G_F)^{-1/2}$	Higgs vacuum value, 246.22 GeV
G_F	`gFermi`	1.1663788×10^{-5} GeV^{-2}	Fermi constant
m_W, m_Z, m_H	`mWb`, `mZb`, `mHb`	80.3692, 91.1880, 125.20 GeV	boson masses (PDG 2025)
θ_W	`cW`, `sW2`	$\cos\theta_W = m_W/m_Z$	weak angle (on shell)
e, α	`eEW`	$e = g\sin\theta_W$, $\alpha = e^2/(4\pi)$	electric coupling (tree level)
M^2_{ij}	`mGB`	Hessian of $\lvert D_\mu\langle\Phi\rangle\rvert^2$ in (W^1, W^2, W^3, B)	gauge-boson mass matrix

$$A_{SU(3)^2U(1)} = \sum_{N_f=3} \chi_f d_f Y_f, \quad A_{SU(2)^2U(1)} = \sum_{d_f=2} \chi_f N_f Y_f, \quad A_{U(1)^3} = \sum_f \chi_f N_f d_f Y_f^3, \tag{3.1}$$
$$A_{\mathrm{grav}^2\mathrm{U}(1)} = \sum_f \chi_f N_f d_f Y_f, \quad A_{SU(3)^3} = \sum_{N_f=3} \chi_f d_f.$$

4. Masses and mixing

symbol	Mathematica	definition	meaning
Y_u, Y_d, Y_e, Y_ν	`yU`, `yD`	complex 3×3	Yukawa matrices
U_L, U_R	`uuL`, `udL`, …	$Y = U_L\,\mathrm{diag}(y)\,U_R^\dagger$	singular value decomposition
V_{CKM}	`vCKM`	$U_{uL}^\dagger U_{dL} =$ (4.1)	quark mixing matrix
U_{PMNS}, U	`uPMNS`	$U_{eL}^\dagger U_{\nu L} =$ (4.1)	lepton mixing matrix
s_{ij}, c_{ij}, δ	`mixU[s12, s13, s23, d]`	$c_{ij} = \sqrt{1-s_{ij}^2}$	mixing angles and phase
J	`jCKM`, `jPMNS`	$\mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$ (likewise for U)	Jarlskog invariant
$\bar\rho + i\bar\eta$, A	`rbCKM`	$-V_{ud}V_{ub}^*/(V_{cd}V_{cb}^*)$, $A = s_{23}/s_{12}^2$	Wolfenstein parameters
$m_u, \dots, m_t$	`mUp`, `mDown`	PDG 2025	quark masses
m_1, m_2, m_3	`mNu`	0, $\sqrt{\Delta m_{21}^2}$, $\sqrt{\Delta m_{31}^2}$ (eV)	neutrino masses
Δm_{ij}^2	`dm21`, `dm31`	$m_i^2 - m_j^2$	mass-squared differences (eV^2)
y_t	—	$\sqrt{2}m_t/v$	top Yukawa coupling

$$U(s_{12},s_{13},s_{23},\delta)=\begin{pmatrix}1&0&0\\0&c_{23}&s_{23}\\0&-s_{23}&c_{23}\end{pmatrix}\begin{pmatrix}c_{13}&0&s_{13}e^{-i\delta}\\0&1&0\\-s_{13}e^{i\delta}&0&c_{13}\end{pmatrix}\begin{pmatrix}c_{12}&s_{12}&0\\-s_{12}&c_{12}&0\\0&0&1\end{pmatrix}\qquad(4.1)$$

5. Cosmology on the brane

symbol	Mathematica	definition	meaning
H_a	—	a'/a	Hubble rate of ordinary space
ρ_4, ε_8	`r4`	$\varepsilon_8=\rho_4/V_\perp$	energy densities in 4 and 8 dimensions
Λ_4	—		cosmological constant of the brane
H_0, Ω_m, Ω_r, Ω_Λ	`h0P`, `omM`, `omR`, `omL`	67.4 km/s/Mpc, 0.315, 9.14×10^{-5}, $1-\Omega_m-\Omega_r$	Planck 2018 parameters
t_0	`ageU`	(5.1)	age of the universe

$$3H_a^2=8\pi G_4\rho_4+\Lambda_4,\qquad t_0=\frac{1}{H_0}\int_0^1\frac{da}{a\sqrt{\Omega_r a^{-4}+\Omega_m a^{-3}+\Omega_\Lambda}}.\qquad(5.1)$$

6. dirac16complex at the brane

symbol	Mathematica	definition	meaning
ψ	`dirac16complex`; `z1B`, `z2B`	16 complex components	the bulk field; random test spinors
γ_μ, γ_5	`g4D`, `g5D`	(6.1)	Dirac representation of the brane
ρ_k	`rhoK`	(6.1)	transverse Clifford matrices, $\rho_k\rho_l+\rho_l\rho_k=-2\delta_{kl}$
ρ_{4567}	`rho4567`	$\rho_4\rho_5\rho_6\rho_7$	transverse chirality, eigenvalues ±1
R_A	`rBr`	(6.2)	brane form of the Dirac matrices
U	`uInt`	$\Gamma_A U=UR_A$, $U^\dagger U=1$	intertwiner (Schur average)
ζ, ψ_a	—	$\psi=U\zeta$, $\zeta=\sum_a\psi_a\otimes e_a$	four brane Dirac spinors
$\bar\psi_a$	`bar4[z, a]`	$\psi_a^\dagger\gamma_0$	brane Dirac adjoint
χ	`χ17`	$i\Gamma_0\cdots\Gamma_7=U(\gamma_5\otimes\rho_{4567})U^\dagger$	eight-dimensional chirality
$\vec q$, $\vec n$	`qKK`	$\vec q=2\pi\vec n/L_c$, $\vec n\in\mathbb{Z}^4$	auxiliary momentum on the torus
X	`xKK`	$\gamma_5\otimes(\vec\rho\cdot\vec q)/b_0$	Kaluza–Klein operator, $X^2=-\lvert\vec q\rvert^2/b_0^2$
$m_{\vec n}$	—	$\sqrt{m^2+\lvert\vec q\rvert^2/b_0^2}$	Kaluza–Klein mass
m	`mPs`; `mPsi`		mass of dirac16complex (7/10 in the KK check; 1 GeV in the portal example)

$$\gamma_0=\mathrm{diag}(1,1,-1,-1),\quad\gamma_i=\begin{pmatrix}0&\sigma_i\\-\sigma_i&0\end{pmatrix},\quad\gamma_5=i\gamma_0\gamma_1\gamma_2\gamma_3,\qquad(6.1)$$

$$\rho_4=i\gamma_0,\ \rho_5=-\gamma_1,\ \rho_6=-\gamma_2,\ \rho_7=-\gamma_3,$$
$$R_\mu=\gamma_\mu\otimes1_4\ (\mu=0,\dots,3),\qquad R_k=\gamma_5\otimes\rho_k\ (k=4,\dots,7),\qquad R_AR_B+R_BR_A=2\eta_{AB}1_{16}.\qquad(6.2)$$

7. Portals

symbol	Mathematica	definition	meaning
y^L_{is}, y^R_{is}	—		Yukawa portal couplings (i flavour, $s = 1, 2$)
$\psi^{(L)}_s$, $\psi^{(R)}_s$	—	brane-right-handed parts of the left/right half of ψ	
P_R	—	$\frac{1}{2}(1+\gamma_5)$	brane right-handed projection
μ_{ia}	`muPortal`	$y_{ia}v/(\sqrt{2}\sqrt{V_\perp})$	portal Dirac mass (eV)
m_D	`mDnu`	$U_{\rm PMNS}\,\mathrm{diag}(m_1, m_2, m_3)$	Dirac neutrino mass matrix
$\mathcal{M}$	`mFull`	$\begin{pmatrix} m_D & \mu \\ 0 & m1_4 \end{pmatrix}$	7×7 Dirac mass matrix of the zero mode
N	`nAct`	upper-left 3×3 block of the left singular vectors of $\mathcal{M}$	non-unitary light mixing matrix
η	—	$N = (1-\eta)U$, $\eta \approx \mu\mu^\dagger/(2m^2)$	deviation from unitarity
$1 - NN^\dagger$	`devNN`	$\approx \mu\mu^\dagger/m^2 = 2\eta$	printed as `(1 - N.N^dag)_ee`
G^f_P, n_ψ, V^f_P	—	$V^f_P = G^f_P n_\psi$	vector portal coupling, density, potential
λ_P	—	mass shift $\lambda_P v^2/(2V_\perp)$	Higgs portal coupling
T_ψ, $T_{\rm stab}$, T_W	—		energy–momentum tensors of ψ, the stabilizing stress, the brane fields
$J_{\rm portal}$	—	$\partial\mathcal{L}_{\rm portal}/\partial\bar\psi$	portal source of the dirac16complex equation

8. Neutrino flavour

symbol	Mathematica	definition	meaning
E, L	`en`, `L`	GeV (MeV in the Sun), km	energy and baseline
$P(\alpha \to \beta)$	`pVac`, `pMat`	(8.1), (8.2)	oscillation probability; `p[[α, β]]`
X_{ij}	`xx` in `xFormula`	$U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}$	quartic product
Δ_{ij}	`phaseF`	$\Delta m^2_{ij}L/(4E)$	oscillation phase
n_e, Y_e, ρ	`rho`, `ye`	$n_e = \rho Y_e N_A$	electron density, electron fraction, mass density
N_A	`NAv`	$6.02214076 \times 10^{23}$	Avogadro's number
V_{CC}	`vCC[rho, ye]`	$\sqrt{2}G_F n_e$	charged-current potential (eV)
H	`h` in `pMat`	(8.2)	flavour Hamiltonian in matter
θ_m	`thM[E, V]`	(8.3)	mixing angle in matter
$R_\odot$	`rSun`	6.957×10^8 m	solar radius
P_{ee}	`peeSunNum`, `peeSunAd`	(8.4)	solar electron-neutrino survival probability
$\rho_{\rm res}$	`rhoRes`	$\Delta m^2 \cos 2\theta/(2EV_{CC}(1\,\mathrm{g/cm^3}))$	MSW resonance density
T_γ, T_ν	`tNu`	$T_\nu = (4/11)^{1/3}T_\gamma$, $T_\gamma = 2.7255$ K	photon and neutrino temperatures
n_ν	`nNu`	$\frac{3\zeta(3)}{2\pi^2}T_\nu^3$	relic neutrino density per flavour

$$P(\alpha \to \beta) = \Big| \sum_i U_{\beta i} U^*_{\alpha i} e^{-i m_i^2 L/(2E)} \Big|^2 = \delta_{\alpha\beta} - 4 \sum_{i>j} \mathrm{Re} X_{ij} \sin^2 \Delta_{ij} + 2 \sum_{i>j} \mathrm{Im} X_{ij} \sin 2\Delta_{ij}, \tag{8.1}$$

$$H = \frac{1}{2E} U \,\mathrm{diag}(0, \Delta m^2_{21}, \Delta m^2_{31})\, U^\dagger + \mathrm{diag}(V_{CC}, 0, 0), \qquad P(\alpha \to \beta) = \left| \left(e^{-iHL}\right)_{\beta\alpha} \right|^2, \tag{8.2}$$

$$\tan 2\theta_m = \frac{\sin 2\theta}{\cos 2\theta - 2EV/\Delta m^2}, \tag{8.3}$$

$$P_{ee} = c_{13}^4 \, \frac{1 + \cos 2\theta_{12} \cos 2\theta_m(0)}{2} + s_{13}^4 \quad \text{(adiabatic)}, \qquad n_e(r) = 245\, N_A\, e^{-10.54 r/R_\odot} \text{ cm}^{-3}. \tag{8.4}$$

9. Symbols that look alike

symbol	**not to be confused with**
W^I_μ (electroweak gauge fields, code `wW1` …)	W (the world brane); w_1, w_2, w_3 (Lovelock couplings of the gravity part, code `w1` …)
γ_μ (4×4, brane)	Γ_A (16×16, 7+1 spacetime)
ρ_k (transverse matrices)	ρ (mass density), $\bar\rho$ (Wolfenstein parameter), ρ_4 (energy density)
U (intertwiner, section 6)	U, U_{PMNS} (lepton mixing matrix), U_L, U_R (singular vectors)
η (metric: the squares of the octonion basis)	η (deviation from unitarity), $\bar\eta$ (Wolfenstein parameter), $\{7, 1, 0\}$ (xAct signature list: seven space-like dimensions, one time-like)
λ (Higgs self-coupling)	λ_P (Higgs portal coupling), $\lambda = s_{12}$ (Wolfenstein)
μ^2 (Higgs mass parameter)	μ_{ia} (portal Dirac mass), μ (muon flavour)
χ_f (chirality sign of a field)	χ (eight-dimensional chirality matrix)
A (Wolfenstein parameter)	$A_{\ldots}$ (anomaly coefficients), $A(\alpha \to \beta)$ (amplitude)
H (flavour Hamiltonian)	H_a (Hubble rate), H_0 (Hubble constant)
e (electric coupling)	e (electron flavour), e_a (unit vectors of $\mathbb{C}^4$)